Random holdout

Temporal holdout

A

B

C

D

All substitutions

De novo only

antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorisation)
Substitution matrix
Where only
Codon accessibility
Uniform

0.86
0.83
0.79
0.79
0.70
0.50
0.32
0.10

0.81
0.77
0.73
0.72
0.65
0.44
0.30
0.10

0.45
0.36
0.45
0.05
0.46
0.17
0.33
0.10

0.51
0.39
0.50
0.05
0.47
0.18
0.34
0.10

0.0
0.4
0.8

Recall at top 10%

All substitutions
De novo only

95% interval excludes zero
interval includes zero

E

F

G

H

Vs single−tree

Vs where x which

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled

0.0
0.2
0.4
0.6
0.8

−0.4
0.0
0.4

−0.2
0.0
0.2
0.4

Multi−tree minus comparator, recall at top 10%